

6(a)(i)	increase in kinetic energy $= \frac{1}{2}mv^2 - \frac{1}{2}mu^2$ $= \frac{1}{2}m(v^2 - u^2)$ $= \frac{1}{2}(1350)(24^2 - 3.0^2)$ $= 382725 \text{ J}$ $\approx 380000 \text{ J (2s.f.)}$
6(a)(ii)	The increase in kinetic energy (KE) of the second car is the same as the increase in KE of the first car. This is because the KE increase depends only on the initial and final velocity. Initial velocity of the 2 cars are the same (3.0 m s^{-1}). Final velocity of the 2 cars are also the same (24 m s^{-1}).
6(b)	Stage 1: Using $s = \left(\frac{u+v}{2} \right) t$ $s_1 = \left(\frac{0+3.0}{2} \right) 1.2$ $= 1.8 \text{ m}$

	Stage 2: Using $v^2 = u^2 + 2as$ $24^2 = 3.0^2 + 2(4.5)s_2$ $s_2 = 63 \text{ m}$ total distance travelled = $s_1 + s_2$ $= 1.8 + 63$ $= 64.8 \text{ m}$ $\approx 65 \text{ m (2 s.f.)}$
6(c)	centripetal force = $m \frac{v^2}{r}$ $= 1350 \left(\frac{24^2}{65} \right)$ $= 11963 \text{ N}$ $\approx 12000 \text{ N (2 s.f.)}$
6(d)(i)	For a car moving forward, the car tyres exert a backward frictional force on the road. By Newton's third law of motion, the road exerts an equal and opposite forward frictional force on the tyres, causing the car to accelerate.
6(d)(ii)	For the car to follow the bend in the road, friction must provide the centripetal force to enable the car to travel in a circle. Otherwise the car would continue travelling in a straight line.
6(e)(i)	maximum frictional force = $0.70 \times \text{weight of car}$ $= 0.70 \times 1350 \times 9.81$ $= 9279.45 \text{ N}$ $< 12000 \text{ N}$ Frictional force provides the centripetal force required to travel around the bend. Since the maximum frictional force (9279.45 N) is less than the centripetal force (12000 N) required for the car to travel around the bend at 24 m s^{-1}, it therefore cannot travel around the bend at the required speed.
6(e)(ii)	The normal contact force acting on the tyres of the car has a horizontal component that acts towards the centre of the circular path. This adds to the horizontal component of the frictional force and hence allows the car to be able to travel around the bend at 24 m s^{-1} .
6(e)(iii)	Wet weather reduces frictional force between the tyres and the road. Hence θ must increase so that the horizontal component of the normal contact force can be increased to compensate for the decrease in frictional force.
7(a)(i)	Background radiation is a measure of the level of ionising radiation present in